Escuela Superior Politécnica del Litoral (ESPOL)
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The Editors
Scientific Reports

Dear Editors,

We submit the manuscript "Re-estimation-verified selection of small parameter subsets for systems biology models" by Mónica Miranda R., Carlos Salazar, César Martín and Adriana Aguirre for consideration as an Article in *Scientific Reports*.

Dynamical models in systems biology routinely contain tens to hundreds of unknown rate constants, far more than the available data can determine. Calibrating all of them is slow, numerically unstable and yields parameter values that the data do not constrain. Existing subset-selection criteria choose a few parameters to estimate, but, to our knowledge, they do not test whether re-estimating the chosen subset actually reproduces what the full model predicts for the measured outputs. Our method closes this gap: starting from published parameter values, it ranks the biological parameters by their effect on the measured outputs, rejects redundant ones, accepts a subset only if re-estimating it (together with the measurement scaling and offset factors) reproduces, within a set tolerance, the response of the full model to random perturbations of all parameters, and removes parameters down to an imposed minimum of two.

We evaluated the method on 34 published models with real data from the PEtab benchmark collection, under a common protocol with null baselines and four established criteria (Brun's collinearity index, Morris screening, profile likelihood and standard pivoted QR). The main findings are:

- In 31 of the 34 problems, re-estimating a subset of usually two biological parameters (the imposed minimum; one sufficed in 15 of 27) together with the calibration parameters reproduced, within the chosen tolerance, the local response of the measured outputs, as confirmed by nonlinear re-estimation; 22 subsets remained admissible up to $\pm 50\%$ in both robustness designs (five of them evaluated to first order beyond $\pm 5\%$).
- With the other parameters at their published values, all selected parameters were practically identifiable in 25 problems (23 with a misfit-scaled threshold; 65 of 74 parameters, by profile likelihood).
- The calibration parameters alone, or random subsets of the same size, passed the test in 9 and 16 of 28 problems; we report this openly because it shows in which problems the test discriminates between subsets.
- Selected from the same training data, our subsets predicted synthetic data generated near the published values more often than pivoted QR and profile likelihood, and not detectably differently from Brun's collinearity index; on real held-out data, no criterion differed significantly from ours.

We have been careful to state the scope of these results. The ranking by sensitivity is not new,

and we say so: the contribution is the verification by re-estimation, the reduction by backward elimination and the explicit reporting of the three models in which no reduction was found. We do not claim that reduced models predict better than full models on data unseen by the published estimates, and we report that the minimum subset size of two is a conservative choice, which we test explicitly. The mathematical statements on which the method relies are formally verified in the Lean 4 proof assistant (32 modules, 309 theorems and lemmas, and the hypotheses on the dynamics of 28 benchmark models at their nominal parameter values); their application to each numerical result is checked numerically, not certified, and the manuscript specifies when each formal statement applies and when it does not.

We believe the work suits *Scientific Reports* because the problem it addresses, too many parameters for the available data, is shared by systems biology, pharmacology and epidemiology, and because its conclusions rest on a large public benchmark and on openly reproducible analyses rather than on a single case study. A reproducibility package (Zenodo, doi:10.5281/zenodo.23338175) contains the code of the method and of every analysis, all stored results, the programs that regenerate every figure and table, and the Lean proofs, with a manual that describes three independent levels of checking.

The study analyses only publicly available, previously published benchmark models and data; no new data from human participants were collected. The main text is somewhat above the recommended length because the comparison with existing criteria and the scope of the formal verification needed to be reported in the main text; detailed results are given in the Methods and in the Supplementary Information.

Suggested reviewers. [Name, affiliation, e-mail and area of expertise of three to five suggested reviewers.]

Excluded reviewers. [Names of any referees to be excluded, or "None".]

Prior discussions. We have had no prior discussions with any Editorial Board Member of *Scientific Reports* about this work.

Related submissions. [State the status of any related conference or journal submission of this work, or "None".]

All authors have approved the manuscript and agree with its submission. The manuscript has not been published and is not under consideration elsewhere. The corresponding author is Mónica Miranda R.

Sincerely,

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